

DC-DC Buck Converter Report

B1-G5

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1 Objective

To design and analyze a DC-DC Buck Converter using a TL494 PWM controller and MOSFET driver circuit.

2 Theory

A buck converter is a high-efficiency DC-to-DC step-down converter that reduces a higher input voltage to a lower output voltage by rapidly switching a transistor (usually a MOSFET) on and off. It uses an inductor, diode, and capacitor to store energy and smoothen out the voltage, with an average output voltage regulated by the duty cycle.

The output voltage is ideally given by:

$$V_o = D \cdot V_{in} \quad (1)$$

where D is the duty cycle. The output of the buck converter corresponds to the above result when we do not take non-idealities into account, on considering them the efficiency of our converter is altered.

3 Driver Circuit

We used the TL494 board, for generating the pulses for our Buck converter. The TL494 PWM controller generates pulses which are fed into the TC4428 MOSFET driver. The driver amplifies the signal and drives the MOSFET.

4 Power Circuit

The power circuit of the DC-DC Buck converter is responsible for stepping down the input DC voltage to a desired lower output voltage using high-frequency switching.

It primarily consists of a power MOSFET, a fast recovery diode, an inductor, an output capacitor, and a load. The MOSFET acts as a high-speed switch controlled by the PWM signal generated from the driver circuit. The diode provides a path for inductor current when the MOSFET is turned OFF.

This circuit operates efficiently due to reduced switching losses and is widely used in power electronic applications such as voltage regulators, battery chargers, and embedded systems.

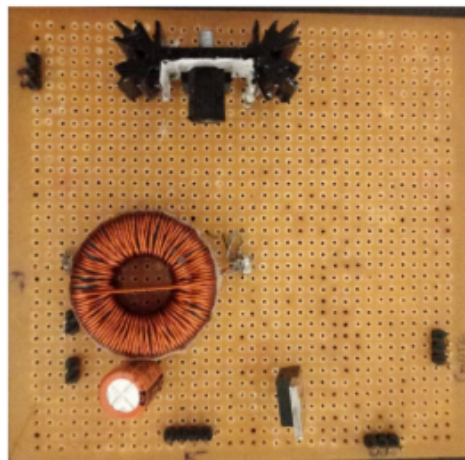


Figure 2: Buck Converter Circuit

Table 2: Power Circuit Components

Qty	Component	Value	Description
1	MOSFET	IRFZ44N	Power switch
1	Diode	Fast recovery	Freewheeling diode
1	Inductor	1 mH	Energy storage element
1	Capacitor	470 μ F, 63V	Output filter capacitor
1	Capacitor	100 μ F, 25V	Output smoothing capacitor
1	Load (Rheostat)	40 Ω , 3A	Variable load
1	Power Supply	± 12 V	Input supply
1	Heat sink	–	MOSFET cooling

5 Working Principle

When the MOSFET is ON, energy is stored in the inductor as current flows through it. When the MOSFET is OFF, the inductor releases its stored energy through the diode to the load, maintaining a continuous current. The output capacitor filters the voltage to reduce ripple and provide a stable DC output, while the inductor smooths the current waveform. The output voltage is controlled by varying the duty cycle of the PWM signal.

6 Efficiency

The efficiency of an Ideal Buck converter (not taking non-idealities into account) is 100% which is $V_{out} \cdot I_{out} = V_{in} \cdot I_{in}$.

7 Graphs



Figure 3: Output current and Output Voltage in CCM.



Figure 4: Inductor Current with respect to gate pulse in CCM.

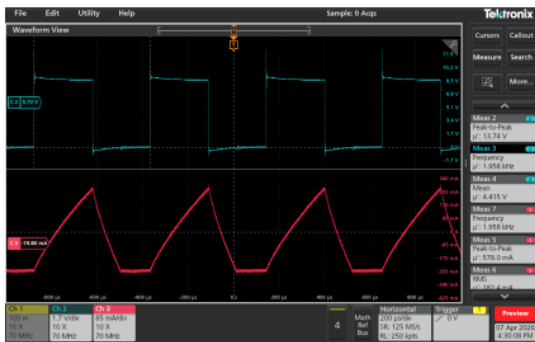


Figure 5: Inductor Current reaching zero in DCM.

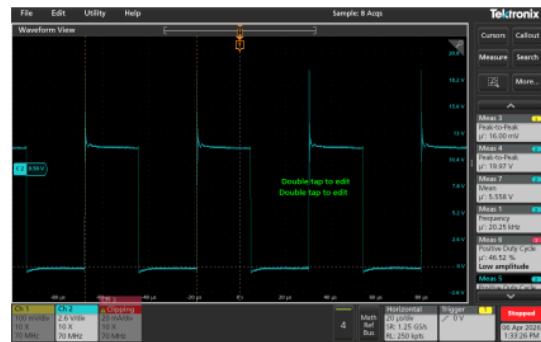


Figure 6: Voltage across the switch in CCM.

8 Conclusion

The Buck converter successfully steps down the input voltage. The output voltage varies more or less linearly with duty cycle, validating the theoretical relation.